

№14:

UNIX

, man

2026-05-14

- UNIX
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- man
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1:

:

- t1
- t2
- mkdir

semaphore.sh

```
#!/bin/bash

t1=${1:-10}
t2=${2:-3}
lockfile=${3:-/tmp/my_resource.lock}

echo "    $$    . t1=$t1 t2=$t2 lock=$lockfile"

waited=0

while [ $waited -lt $t1 ]; do
    if mkdir "$lockfile" 2>/dev/null; then
        echo "    $$    (    $waited    .)"

        sleep $t2

        echo "    $$    "
        rmdir "$lockfile"

        exit 0
    fi

    echo "    $$    ... ($waited/$t1)"

    sleep 1
    waited=$((waited+1))
done

echo "    $$    $t1    ."
exit 1
```

semaphore.sh

1

```
tty
/dev/pts/1
```

```
./semaphore.sh 15 5
```

:

```
1234 . t1=15 t2=5 lock=/tmp/my_resource.lock
1234 ( 0 .)
1234
```

2

```
tty
/dev/pts/2
```

```
./semaphore.sh 15 5 > /dev/pts/1 2>&1 &
```

:

```
[1] 1235
```

-
-
-

mkdir

The screenshot shows two terminal windows. The left window, titled 'sway', shows the command `./semaphore.sh 15 5` being executed. The right window, titled 'hhl@LAPTOP-658L8DAA: ~/work/study/2026-2027/Операционные системы/os-intro/labs/Lab14', shows the command `./semaphore.sh 15 5 > /dev/pts/0 2>&1 &` being executed. Below the command in the right window, the output of the process is shown, including the creation of the lock file, the acquisition of the resource by process 3684, and the subsequent acquisition and release of the resource by process 3689.

```
(base) hhl@LAPTOP-658L8DAA:~$ cd ~/work/study/2026-2027/Операционные системы/os-intro/labs/Lab14
(base) hhl@LAPTOP-658L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/Lab14$ nano semaphore.sh
(base) hhl@LAPTOP-658L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/Lab14$ chmod +x semaphore.sh
(base) hhl@LAPTOP-658L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/Lab14$ tty
/dev/pts/0
(base) hhl@LAPTOP-658L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/Lab14$ ./semaphore.sh 15 5
Процесс 3684 запущен. t1=15 t2=5 lock=/tmp/my_resource.lock
Вывод направлен на терминал /dev/pts/0
Процесс 3684 получил ресурс (ожидал 0 сек.)
Процесс 3684 освобождает ресурс
(base) hhl@LAPTOP-658L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/Lab14$
(base) hhl@LAPTOP-658L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/Lab14$ ./semaphore.sh 15 5 > /dev/pts/0 2>&1 &
[1] 3689
(base) hhl@LAPTOP-658L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/Lab14$
```

2: man

man:

- - /usr/share/man/man1
 - zless
-

myman.sh

```
#!/bin/bash

if [ $# -ne 1 ]; then
    echo "      : $0      "
    exit 1
fi

manpage="/usr/share/man/man1/${1}.1.gz"

if [ -f "$manpage" ]; then
    zless "$manpage"
else
    echo "      '$1'      /usr/share/man/man1"
    exit 1
fi
```

myman.sh

```
./myman.sh ls
```

:

●

```
./myman.sh nonexistent_command
```

$$\vdots$$

```
'nonexistent_command'      /usr/share/man/man1
```

```
(base) hnt@LAPTOP-05L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab14$ ./myman.sh notexist
Справка по 'notexist' не найдена в /usr/share/man/man1
```

3:

:

-
- \$RANDOM
-

random_letters.sh

```
#!/bin/bash

length=${1:-20}

letters="abcdefghijklmnopqrstuvwxyz"
result=""

for ((i=0; i<length; i++)); do
    idx=$((RANDOM % 26))
    result="${result}${letters:idx:1}"
done

echo "          : $result"
```

random_letters.sh

```
./random_letters.sh
```

:

```
: kxmzowpqrbcvntghgejdsa
```

30

```
./random_letters.sh 30
```

:

```
: plmoknijbuhvygctfxrdzeswaqlmn
```

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab14$ ./random_letters.sh 30
Случайная последовательность: tarubztruxnfyrpbnudvdihaxmhunl
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab14$ ./random_letters.sh
Случайная последовательность: sjrmreqnfgzuawhphzgp
```

1. while

:

```
1 while [ $1 != "exit" ]
```

:

```
while [ "$1" != "exit" ]
```

2.

```
str="Hello"
str2="World"

result="$str $str2"
result="{str}{str2}"
```

3. **seq**

- echo {1..10}
 - for
 - while
- :

```
for ((i=1; i<=10; i++)); do
    echo $i
done
```

4.

```
$((10 / 3))
```

:

```
3
```

5. zsh bash

-
- glob-
- 1
-

6. for

:

```
for ((a=1; a <= LIMIT; a++))
```

:

```
for ((a=1; a <= LIMIT; a++)); do
```

```
done
```

7. Bash:

Linux-

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-
- bash-
- \$RANDOM
- -
- mkdir
- man

UNIX.

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